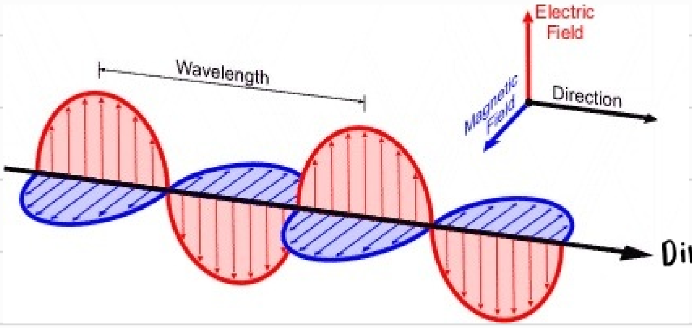
Electromagnetic (EM) waves

$\boxed{v = f\lambda \Rightarrow c = f\lambda}$ $\boxed{c = \frac{|E|}{|B|}}$

RTR



Direc. of Propagation / Direc. of energy flow [Poynting Vector, $\vec{S}$]

Poynting Vector, $\boxed{\vec{S}(x,y,z) = \frac{1}{\mu_0}\left[\vec{E}(x,y,z,t) \times \vec{B}(x,y,z,t)\right]}$

Magnitude of Poynting Vector, $|\vec{S}(t)| = \frac{1}{\mu_0}\left[E(t)B(t)\sin\theta\right]$ Angle btwn E & B [Always 90°]

$$\boxed{|\vec{S}(t)| = \frac{1}{\mu_0}\left[E(t)B(t)\right]}$$

Since the unit for E is V/m, B is Newton per meter per ampere [N/m/A]

$|\vec{S}(t)|$ is the Rate of Energy Flow per meter², [W/m²] at a given instant
(Power per Area)

In other words, $|\vec{S}(t)|$ is the instantaneous power of the EM wave

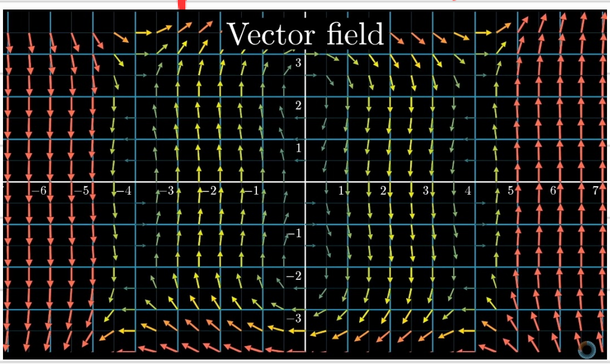
However, since EM waves oscillates crazily fast, it will useful to calculate the average Intensity of the EM wave over a period of time

$$\boxed{I_{avg} \text{ over time } t = S_{avg}}$$
$$= \frac{1}{t}\int_0^t |\vec{S}(t)|\, dt$$

* Take note!

$\vec{E}(x,y,z,t)$ and $\vec{B}(x,y,z,t)$ aren't just singular vectors, they are vector fields.
Vector fields associate each point in space with a vector [some magnitude & direc.]

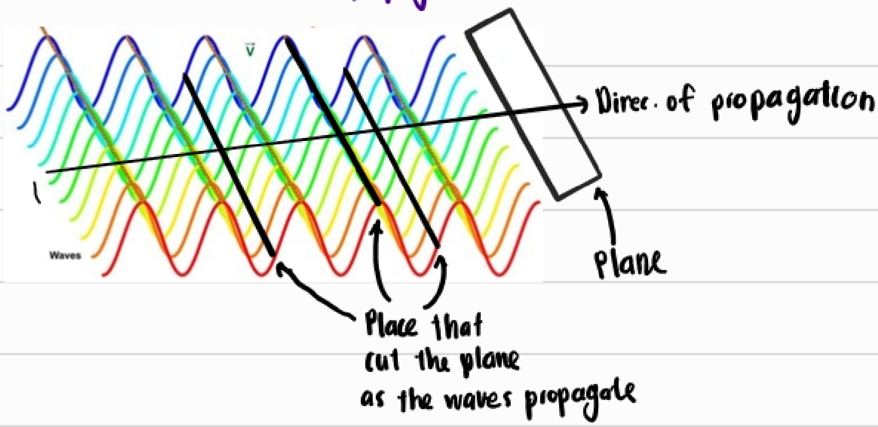
For example:



Sinosuidal Plane EM waves

Sinosuidal ~ waves oscillate fllwing a sine wave pattern

Plane waves ~ The magnitude and orientation is same within each plane perp. to the direc. of propagation



[Notice that waves oscillate in-phase within the plane]

∴ A sinosuidal plane wave is an EM wave where:

- If u take a plane perp. to direc. of propagation, the electric field vectors, $\vec{E}(x,y,z,t)$ at all points on that plane ...
  1. Have the same magnitude
  2. Same orientation
  3. Oscillate in phase [Increase OR decrease tgt as time passes]

1, 2, 3 Applies to $\vec{B}(x,y,z,t)$.

*Take Note:

$\vec{E}(x,y,z,t)$ not neccesaraly have the same magnitude as $\vec{B}(x,y,z,t)$.

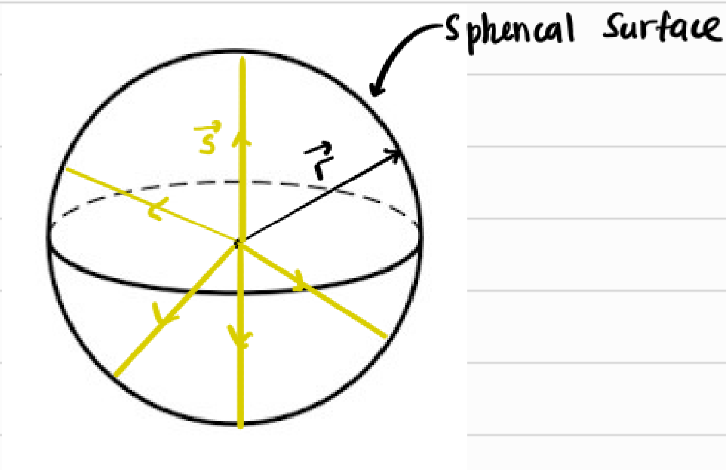
Now since $\vec{E}$ and $\vec{B}$ oscillating harmonically [Repetitive, periodic motion]

$$\Longrightarrow \frac{|E_{max}|}{|B_{max}|} = \frac{E_0}{B_0} = c = \frac{E}{B}$$

And for a sin. plane wave, the fllwing must be true.

$$\boxed{\text{Intensity} = S_{avg} = \frac{E_0^2}{2\mu_0 c} = \frac{(cB_0)^2}{2\mu_0 c}}$$

Sinusoidal spherical EM wave



At each instant,

- $\vec{E}$ is the same
- $\vec{B}$ is the same

Since $\vec{S}$ is radial [parallel w $\vec{r}$], and since $\vec{B}$ & $\vec{E}$ is perp to $\vec{S}$

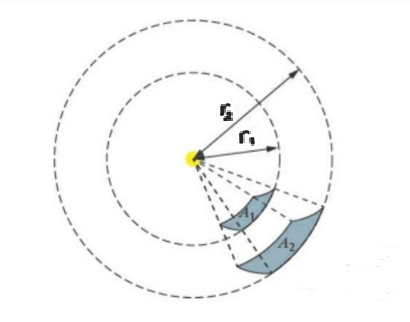
↳ $\vec{B}$ & $\vec{E}$ perp. to $\vec{r}$

↳ $\vec{B}$ & $\vec{E}$ tangent to the surface

Assuming that the avg Power, P_{avg} is uniformly distributed through the surface,

$$\boxed{\text{Average Intensity}, I_{sphere} = \frac{P_{avg}}{\text{Area}} = \frac{P_{avg}}{4\pi r^2}}$$

$I_{spherical} \approx I_{plane}$ when r_{sphere} is large



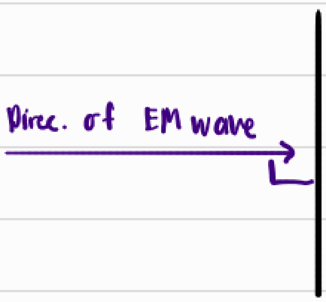
As $r \uparrow$, the area of the spherical surface approaches the area of a plane.

$$\therefore \boxed{I_{spherical} \approx I_{plane}}$$

$$\frac{P_{avg}}{4\pi r^2} \approx \frac{E_0^2}{2\mu_0 c}$$

$$\therefore \underbrace{E}_{\text{Amplitude}} \propto \frac{1}{r}$$

Radiation Pressure, P_R



If wave is totally absorbed by surface,

$$\boxed{P_R = \frac{I}{c}}$$

If wave is totally reflected by surface,

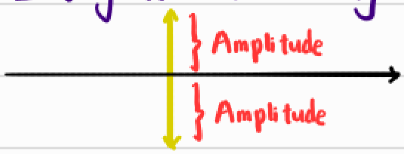
$$\boxed{P_R = 2P_R = 2\left(\frac{I}{c}\right)}$$

Direc. of $\vec{E}$

Polarization of EM wave

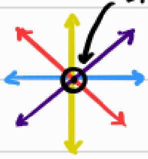
Linearly polarized EM wave, LP

↳ $\vec{E}$ only oscillates along a straight line

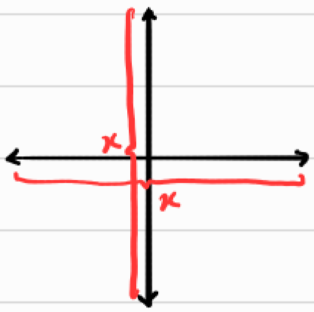


Unpolarized EM wave, UP

↳ Consists of multiple Linearly polarised EM wave uniformly distributed over 360°



All these linearly polarised EM can be resolved into horizontal & vertical linearly polarised EM. [Resultant UP] Simplified UP



Both x & y component have the same amplitude thus same length bcuz the linearly polarised EM wave is uniformly distributed over 360°

Polarization

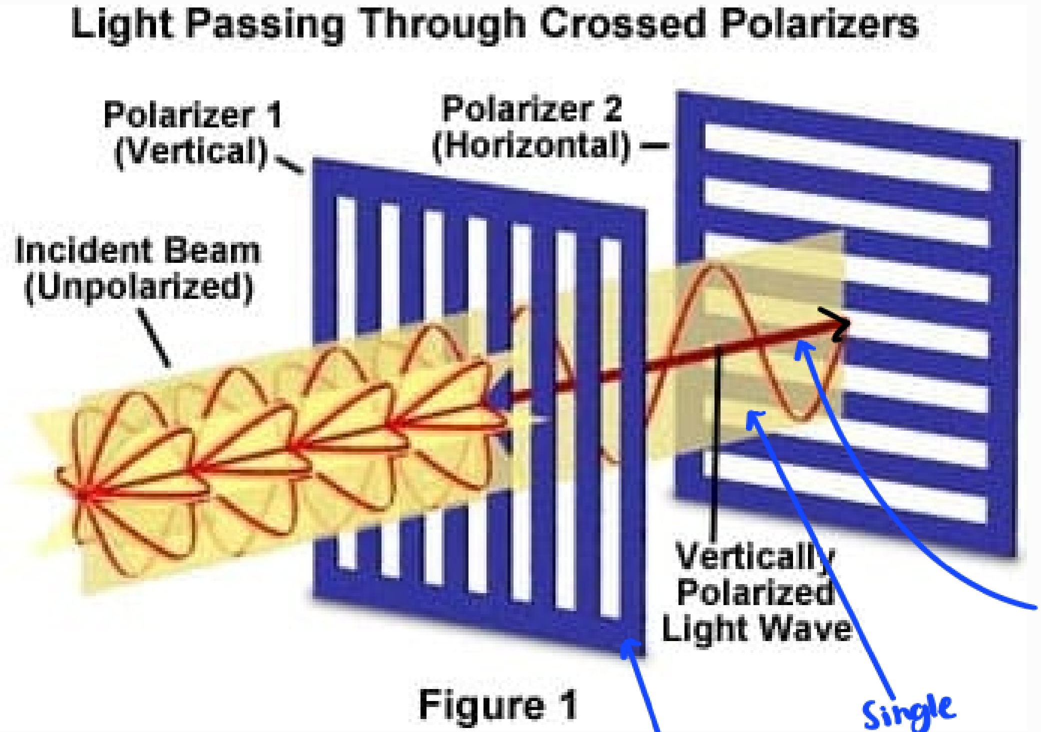


Figure 1

Direction of propagation

Single Plane [Vertical]

Transmission Axis [Plane polarisation]

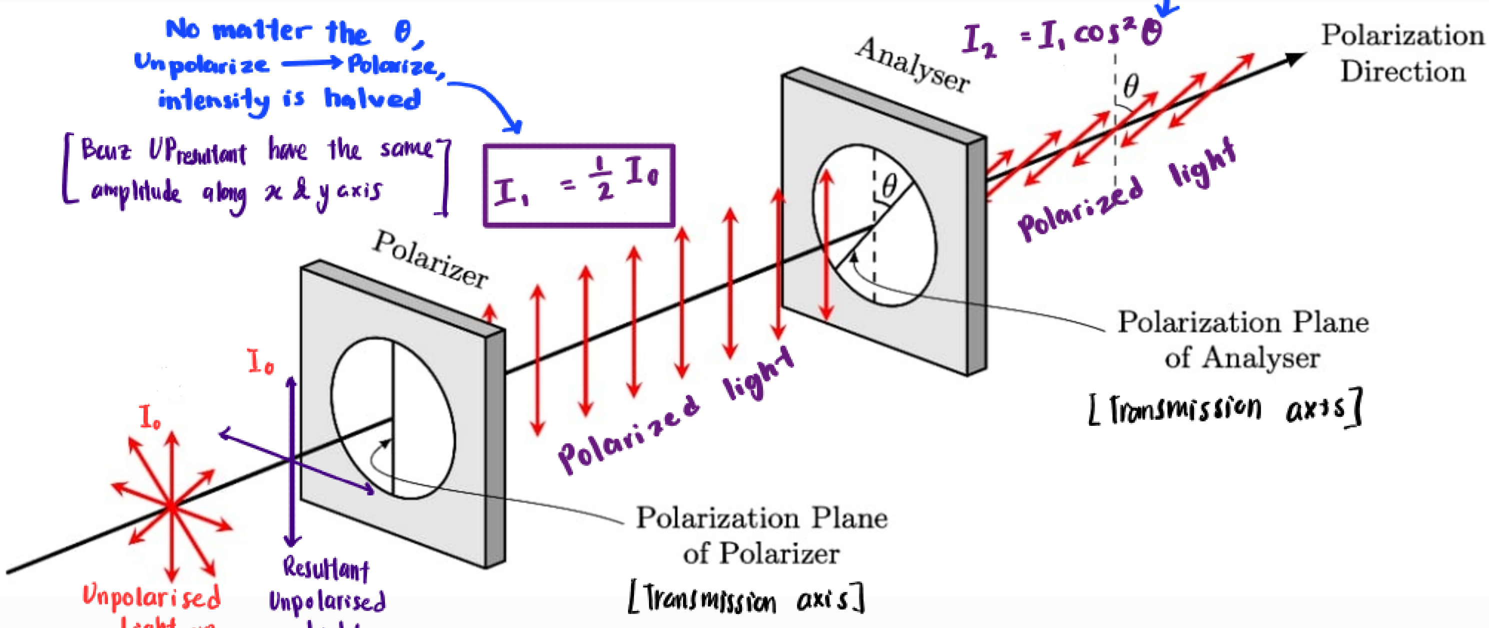
Malus Law

Angle btwn slit of polarizer 1 & polarizer 2
[Angle btwn incident LP & transmission axis]

$I_2 = I_1 \cos^2\theta$

No matter the θ, Unpolarize → Polarize, intensity is halved
[Bcuz $UP_{resultant}$ have the same amplitude along x & y axis]

$\boxed{I_1 = \frac{1}{2} I_0}$



Polarized light

Polarized light

[Transmission axis]

[Transmission axis]

I_0

I_0

Unpolarised Light, UP

Resultant Unpolarised Light, $UP_{resultant}$